# Debate-Based Career Guidance Agent with Complementary Personas

A Dialectic Multi-Agent Architecture for Responsible Algorithmic Career Advisory

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## ABSTRACT

Large Language Models (LLMs) deployed as automated decision-support systems in high-stakes domains—such as career guidance and workforce transition—suffer from three catastrophic systemic failure modes: (1) *The Sycophancy Trap*, wherein models affirm unrealistic candidate aspirations to maximize conversational agreement; (2) *The Illusion of False Certainty*, wherein models present brittle single-path trajectories without calibrated awareness of labor market volatility; and (3) *Latent Demographic Steering*, wherein historical training biases steer underrepresented candidates into low-agency credential traps. To resolve these failures, this paper presents the design, mathematical formulation, and empirical evaluation protocol for a **Dialectic Multi-Persona Career Guidance Architecture**. Deliberation is operationalized via three antagonistic advising personas: The Pragmatic Traditionalist ($\pi_{trad}$, T=0.3, corporate stability and DSA fundamentals), The Entrepreneurial Builder ($\pi_{builder}$, T=0.7, startup speed and AI SaaS shipping), and The Deep-Tech Specialist ($\pi_{spec}$, T=0.4, low-level Linux systems and 5NF database normalization). Personas are orchestrated by a neutral Moderator Synthesizer ($\Phi$) in a 5-stage deliberation loop grounded on a 420-node, 890-edge ACM/IEEE CS2023 prerequisite DAG in Neo4j, dense FAISS labor market retrieval, and DeBERTa-v3 Cross-Encoder NLI factual verification. A composite **Calibrated Epistemic Uncertainty Indicator** $\hat{U}(\blacksquare^{*})$ combining inter-agent Shannon entropy, ontological graph reachability distance, and NLI entailment deficit activates an interactive **Cognitive Forcing Gate** ($\theta_{unc}$ = 0.65) requiring human intervention. In a 500-pair counterfactual benchmark across six ablation tiers, the proposed system elevates Perspective Coverage Rate to 88.4%, achieves an 84.6% Contradiction-Resolution Rate, reduces Unsupported-Claim Rate from 31.4% to 3.2%, satisfies counterfactual demographic parity ($\Delta_{parity}$ = 0.024 ≤ 0.05), and validates salary equivalence under Two One-Sided Tests (TOST, $p < 0.01$) within a ±$2,500 bound.



***Keywords:*** *Responsible AI, Multi-Agent Debate, GraphRAG, Knowledge Graphs, Epistemic Uncertainty, Algorithmic Fairness, Cognitive Forcing.*


## 2. Introduction

**2.1 Socio-Economic Significance:** Career trajectory selection in computing and technology is a high-stakes, multi-objective, and economically irreversible decision. A transition plan impacts tuition expenditures, opportunity costs, and long-term socio-economic mobility. Unlike conversational toys, career guidance systems directly operate at the intersection of human dignity and economic survival.

**2.2 Contemporary Failure Modes:** Monolithic LLM advisory architectures manifest three structural failure modes:
• **The Sycophancy Trap:** RLHF alignment leads models to agree with unrealistic candidate plans (e.g., non-technical background to Staff ML Engineer in 3 months), glossing over fatal prerequisite deficits.
• **The Illusion of False Certainty:** Autoregressive decoding outputs a single smoothed, authoritative roadmap that hides labor market volatility and completion risks.
• **Latent Demographic Steering:** Embedded training biases steer minority and non-traditional candidates toward lower-tier support roles while reserving high-prestige systems roles for traditional profiles.

**2.3 Formal Problem Formulation:** We define the career deliberation environment as the tuple: $\mathbf{M = \langle P, G, D_{rag}, \Pi, T_{debate}, \Phi \rangle}$, where $P$ represents the candidate profile vector, $G$ is the Neo4j prerequisite DAG, $D_{rag}$ is the dense external syllabus/wage corpus, $\Pi$ is the triadic persona ensemble ($\pi_{trad}$, $\pi_{builder}$, $\pi_{spec}$), and $\Phi$ is the neutral Moderator.

## 3. Literature Survey

Our work synthesizes four major research traditions: (1) multi-agent debate for factuality suppression (Du et al., ICML 2024); (2) persona-driven divergent exploration (Liang et al., EMNLP 2023); (3) Responsible AI and automation bias mitigation via cognitive forcing functions (Buçinca et al., CSCW 2021; Cummings, 2004); and (4) hybrid GraphRAG combining structured DAG ontologies with dense vector retrieval (Edge et al., Microsoft Research 2024).

| *Citation / Benchmark* | *Debate?* | *DAG Grounding?* | *Calibrated Uncertainty?* | *Cognitive Gate?* | *Fairness Audit?* | *Local Privacy?* |
|---|---|---|---|---|---|---|
| Du et al. (ICML '24) | Yes (Homog.) | No | No | No | No | Cloud API |
| Liang et al. (EMNLP '23) | Yes (Personas) | No | No | No | No | Cloud API |
| Sharma et al. (ICLR '24) | No (Diagnostic) | No | No | No | No | Cloud API |
| Geyik et al. (KDD '19) | No (Statistical) | No | No | No | Yes (Demog.) | Proprietary |
| Buçinca et al. (CSCW '21) | No (Single) | No | No | Yes (Manual) | No | Empirical |
| Edge et al. (MS '24) | No (Summarizer) | Entity Graph | No | No | No | Cloud API |
| **Proposed System** | **Yes (Triad)** | **Yes (CS2023)** | **Yes (U ∈ [0,1])** | **Yes (Dispute)** | **Yes (TOST)** | **Yes (Local 14B)** |

## 4. Architecture Diagram & System Formulation

**4.1 Architectural Pipeline:** Deliberation executes via a local LangGraph state machine. Candidate profile inputs are stripped of protected demographics (which are isolated in an auditing vault). The three advising personas independently retrieve grounded facts from a 420-node Neo4j prerequisite DAG and dense FAISS vector indices, formulating structured arguments under divergent decoding temperatures.

**SYSTEM ARCHITECTURE FLOW & DATA PIPELINE**

[Candidate Profile P] ■■> [Demographic Isolation Vault] ■■> [LangGraph Deliberation State]
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[Neo4j Prerequisite DAG] (420 Nodes, 890 Edges, ACM/IEEE CS2023 Guidelines) ■■> Cypher Checks
[FAISS Dense Vector DB] (Accredited Syllabi & BLS Quarterly OEWS Wage Data) ■■> Semantic Match
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**[5-STAGE DELIBERATION LOOP]**
Stage 1: Context Intake & Competency Vector Initialization
Stage 2: Parallel Grounded Openings ($\pi_{trad}$ T=0.3, $\pi_{builder}$ T=0.7, $\pi_{spec}$ T=0.4)
Stage 3: Moderator Cross-Exam & Friction Probe Extraction ($F_t$ = ExtractContradictions)
Stage 4: Grounded Rebuttals, Claim Auditing & Concessions
Stage 5: Hybrid Roadmap Synthesis & Dispute Tree (■$_{dispute}$) Extraction
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**[RESPONSIBLE AI & COGNITIVE FORCING ENGINE]**
DeBERTa-v3 Cross-Encoder NLI Entailment Check [P > 0.80] + BLS Wage Validator [p25-p75]
Composite Epistemic Uncertainty: **Û(■$^{*}$) = 0.40·■(Π) + 0.35·D■(G,P) + 0.25·(1 - P■$_{NLI}$)**
If Û > 0.65 OR Disputed Trade-offs ≥ 1 ■■> **[MANDATORY HUMAN OVERRIDE GATE]**
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[Interactive Streamlit Interface: Dispute Tree & Explicit Trade-off Selection] ■■> [Final Roadmap]

**4.2 Prerequisite DAG Derivation:** Rather than using unstructured skill lists, prerequisite edges were formally derived from the **ACM/IEEE CS2023 Curricular Guidelines** across Algorithms (CS-AL), Operating Systems (CS-OS), Data Systems (CS-DS), and Software Engineering (CS-SE). Topological sorting and Cypher reachability queries deterministically eliminate prerequisite hallucinations.

**4.3 Epistemic Uncertainty & Decision Friction:** The Composite Uncertainty Indicator $\hat{U}(\blacksquare^{*})$ synthesizes debate entropy $\blacksquare(\Pi)$, graph distance $D\blacksquare(G,P)$, and mean NLI entailment deficit $(1 - P\blacksquare_{NLI})$. Setting $\theta_{unc} = 0.65$ forces human intervention on approximately 38% of queries, systematically disrupting automation complacency without causing user fatigue.

## 5. Experimental Setup

**5.1 Controlled 6-Tier Ablation Ladder:** We evaluate our system against five controlled baselines:
1. *Baseline A (Monolithic Zero-Shot LLM):* Standard single-prompt Qwen-2.5-14B.
2. *Baseline B (Compute-Matched Self-Consistency):* Majority-vote across k=5 independent sample paths matching compute tokens.
3. *Baseline C (Single-Agent Multi-Perspective Prompting):* A single LLM prompted to analyze Corporate, Startup, and Deep-Tech perspectives sequentially in one context window.
4. *Baseline D (Single-Agent GraphRAG):* Neo4j + FAISS retrieval without multi-agent debate.
5. *Baseline E (Unmoderated Triad):* Triadic personas debating without Moderator synthesis or dispute trees.
6. *Proposed Full System:* Triad + Moderator + CS2023 GraphRAG + NLI Verifier + Cognitive Forcing Gate.

**5.2 Counterfactual Dataset & Metrics:** Evaluated over 500+ synthetic profile pairs (1,000 unique profiles) where skills, coursework, and aspirations are held identical while gender markers, socioeconomic proxies, and college pedigree are varied. Metrics include Perspective Coverage Rate (PCR ≥ 80%), Semantic Contradiction-Resolution Rate (S-CRR ≥ 80%), Unsupported-Claim Rate (UCR < 5%), Counterfactual Parity ($\Delta_{parity} \leq 0.05$), and Two One-Sided Tests (TOST) salary equivalence ($\delta = \pm\$2{,}500$, $p < 0.05$). Inference runs locally on an NVIDIA RTX 4090 GPU (4-bit AWQ, 85 tok/s, 9.4 GPU hours for pilot).

## 6. Results and Discussion

| *Configuration* | *PCR (↑)* | *S-CRR (↑)* | *UCR (↓)* | *$\Delta_{parity}$ (↓)* | *TOST Equiv.* | *ECE (↓)* | *Counselor Likert* |
|---|---|---|---|---|---|---|---|
| Baseline A (Monolithic Zero) | 41.2% | 28.1% | 31.4% | 0.284 | p = 0.412 (Fail) | 0.384 | 2.14 / 5.0 |
| Baseline B (Self-Consistency) | 48.6% | 33.4% | 27.2% | 0.241 | p = 0.358 (Fail) | 0.321 | 2.62 / 5.0 |
| Baseline C (Multi-Persp CoT) | 54.3% | 41.2% | 18.7% | 0.186 | p = 0.245 (Fail) | 0.264 | 3.18 / 5.0 |
| Baseline D (Single GraphRAG) | 51.8% | 44.6% | 4.8% | 0.142 | p = 0.180 (Fail) | 0.198 | 3.65 / 5.0 |
| Baseline E (Unmoderated Triad) | 79.4% | 62.5% | 14.2% | 0.082 | p = 0.084 (Fail) | 0.162 | 3.82 / 5.0 |
| **Proposed Full Architecture** | **88.4%** | **84.6%** | **3.2%** | **0.024** | **p = 0.003 (Pass)** | **0.088** | **4.68 / 5.0** |

**6.1 Core Findings:** As demonstrated in Table 2, Baseline C fails due to *autoregressive context homogenization*: prompting a single model with three perspectives yields only 54.3% PCR and 41.2% S-CRR as attention heads smooth out internal contention. Physical multi-agent separation with asymmetric decoding temperatures preserves authentic dialectic friction (88.4% PCR, 84.6% S-CRR). ACM/IEEE CS2023 GraphRAG compresses Unsupported Claims from 31.4% to 3.2%. Demographic parity distance collapses to 0.024, and salary recommendations pass TOST equivalence (p = 0.003). In double-blind counselor testing, the proposed system achieved a high-risk transition detection AUC of 0.86.

**6.2 Worked Case Walkthrough ('Priya'):** When presented with a 3rd-year student considering a $15,000 predatory bootcamp to become a Staff ML Engineer, the Traditionalist vetoed the plan citing CS-OS deficits; the Specialist exposed the lack of CUDA/memory systems depth; and the Builder redirected the student to open-source RAG micro-SaaS deployments. With calculated uncertainty $\hat{U} = 0.741 > 0.65$, the Cognitive Forcing Gate engaged, presenting an interactive Dispute Tree that averted $15,000 in unaccredited debt and empowered student critical agency.

**6.3 Technical Limitations:** We acknowledge four boundaries: (1) shared base model priors in Qwen-2.5-14B; (2) prerequisite graph curation bounded to 420 computing nodes; (3) decision friction fatigue under high volume; and (4) initial heuristic uncertainty weighting.

## 7. References

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